Chapter 1: Measuring and Describing Data

Statistics for Environmental Science

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Rendered October 22, 2026

This PDF was rendered on October 22, 2026. The book is updated during the semester; the latest version of this chapter is at https://malloryb.github.io/statistics_E538/01-Data_Description.html.

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> *To consult the statistician after an experiment is finished is often merely to ask him to conduct a post mortem examination. He can perhaps say what the experiment died of.*
>
> Sir Ronald Fisher

This chapter covers where data come from and how to describe them. It starts with why statistics matters for environmental questions and how a concept becomes a measurement, then covers scales of measurement and study designs, and ends with the plots and summaries that describe one variable.

The role of statistics in environmental science

Environmental questions are usually questions about change. Is this stream warming? Did the restoration work? They are hard to answer because the systems change on their own even when nothing is done to them. Streams warm and cool over a single day, and vegetation may look the same for years and then suddenly recover. Something is always changing, so finding a real effect means separating it from the change that would have happened anyway.

That's what statistics is for. It gives you a way to say how much of what you observed is the effect you were looking for and how much is ordinary variation, and how confident you should be in telling the two apart. Most environmental claims you will read or make depend on that distinction.

Why numbers matter

Common sense is useful, but it is prone to bias. When we already believe something to be true, we are more likely to interpret new evidence as supporting it, even if it does not.

For example, if you're convinced that industrial farming is the main driver of bee declines, you might see every new drop in bee abundance as confirmation of that belief. Maybe you're right, but maybe disease or weather are stronger contributors. Statistics gives you tools to test these ideas systematically, so that the data decide between farming, disease, and weather instead of your expectations.

Sheer volume is the other reason numbers matter. Satellites measure surface temperature, field stations log rainfall, and sensors track air quality, so a single study can generate thousands of rows. Those rows also vary, because river flow changes by the hour and human activity adds more variability on top. Statistics gives us ways to summarize those thousands of rows, and the variation in them, with a few numbers and plots.

You might ask why you need to learn this yourself instead of handing the analysis to someone who already knows it. There are three reasons.

1. **Design and analysis go together.** A good study starts long before you run a statistical test. If you want to study how fertilizer affects crop yields, the fields you choose to sample decide which comparisons your analysis can make. If the design is poor, like measuring only in unusually wet fields, no analysis can rescue it.

2. **You need statistics to read the science.** Scientific papers on climate change, biodiversity, or pollution report their findings as statistical results, such as a difference between means with a confidence interval or a p-value (later chapters define both), and you need to know what those numbers mean to interpret the paper.
3. **You cannot hire a specialist for every question.** A working knowledge of statistics makes you more self-sufficient, whether you end up working as a scientist, an environmental manager, or a decision-maker who has to evaluate someone else's analysis.

   We are drowning in information, but we are starved for knowledge.

   John Naisbitt, *Megatrends* (1982)

Statistics is useful outside research and outside environmental science. Weather forecasts, air quality alerts, and wildlife population trends are all communicated through numbers, and so are the statistical claims that turn up in news coverage and policy debates. Once you can tell a correlation (Chapter 2) from a cause, or spot a sample too small for the claim it carries, you can check whether the numbers back up what is being said.

Measurement and operationalization

Every dataset begins with measurement. Measurement is assigning numbers, labels, or categories to aspects of the world so they can be recorded and analyzed. Science often deals with broad ideas that need to be pinned down before they can be measured. Ideas like *soil health* or *forest change* are meaningful but vague, so before we can analyze one we have to decide exactly what we mean by it and how to capture it. That process, turning an abstract concept into a specific, measurable procedure, is called **operationalization**. Its result, the specific quantity chosen to stand for the concept, is the **operational definition**.

Two more terms describe the steps that follow it:

- A **measure** is the tool or method used to make observations.
- A **variable** is what you record for each observation once the measure is applied. Its recorded values are the data that end up in the dataset.

Example:

- Concept of interest: *Soil health*
  - **Operational definition:** concentration of organic carbon in the top 30 cm
  - **Measure:** laboratory analysis of soil samples
  - **Variable:** recorded carbon concentration values for each plot

Soil health is the idea we care about, but you cannot go outside and measure an idea. Organic carbon is a defensible stand-in, since healthy soils generally store more of it, but soil health also involves compaction, pH, microbial activity, and how well the soil holds water. So the first decision is which facet of soil health will represent the whole:

Which facet?

- ☐ pH
- ☐ compaction
- ☒ *organic carbon*
- ☐ microbial activity
- ☐ water-holding capacity

Choosing carbon leaves the other facets unmeasured, and it still leaves more choices to make. How deep will you sample? Thirty centimeters is conventional, but 10 cm or a full meter would give different answers:

Sampling depth?

- ☐ 10 cm
- ☒ *30 cm*
- ☐ 1 m

Many more choices follow, and at each one another reasonable person might choose differently. Four of them are listed below.

- **Which carbon counts?** ☒ *organic* ☐ carbonate minerals ☐ total
- **How many cores per plot?** ☐ 1 ☐ 3 ☒ *6* ☐ 9
- **Where in the plot?** ☐ wherever is easiest ☒ *stratified random* ☐ along a fixed transect
- **When in the season?** ☐ spring ☒ *after harvest* ☐ same calendar date each year

Every checked box narrows what "soil health" now means in your dataset, and every unchecked box is a part of soil health your data no longer measure. Another researcher could work down the same list, check different boxes, report a different number, and both of you would be right, because you measured different things and gave them the same name.

Figure 1 collects those choices into one path, from the idea we care about to the number that ends up in the dataset. The faint branch on the left lists the parts of soil health that this dataset no longer measures.

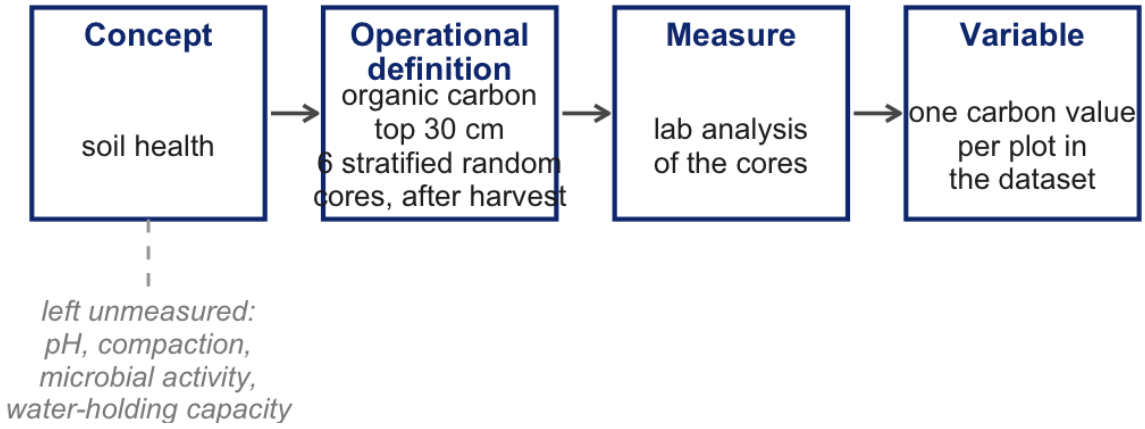


Figure 1: From concept to variable for the soil health example. Each box narrows the one before it, and the facets of soil health that were not chosen drop out at the first step.

There is no single correct way to operationalize a concept. The best choice depends on the question you are asking and how the data will be used. A study of carbon storage has good reason to measure organic carbon, while a study of erosion would more likely measure compaction or water-holding capacity. Even in fields with common practices, each project takes its own judgment. Some principles apply across studies: be precise about what you mean, how you will measure it, and what values are possible.

Scales of measurement

The previous section called the outcome of a measurement a variable. Variables come in different types, and the type determines which statistical tools make sense. You can average daily temperatures, for example, but you cannot average species. A variable's **scale of measurement** says which comparisons its values

support, such as putting them in order, subtracting them, or forming ratios. The four classic scales are nominal, ordinal, interval, and ratio (Stevens 1946).

Nominal scale

A **nominal** variable, also called a **categorical** variable, has values that are names with no order. It makes no sense to average them.

Example. Species. A wildlife survey might record each animal it detects as a green frog, a brook trout, an American crow, or a river otter. None of those is "greater" than another, so they cannot be put in a meaningful order, and there is no such thing as an average species.

A computer will still do the arithmetic, though. Field datasheets often record species as numeric codes, such as 1 for green frog and 4 for river otter, so the computer will average the codes and return a result. Figure 2 shows what comes back.



Figure 2: Averaging the numeric codes of four species returns a number, but not a species. No species has the code 2.5. Icons from Font Awesome Free (CC BY 4.0).

You get a number back, but no species has that code, and it does not stand for any species. Species have no order and no amount, so an average of them is not a species. A variable is nominal when its values are labels, not quantities.

The codes exist only because someone chose numbers for a datasheet. Letters would have worked just as well and left nothing to average. Most nominal variables, like watershed name or trap type, have no numbers attached in the first place.

Ordinal scale

An **ordinal** variable has values with a meaningful order, but the spacing between values is not defined. You can rank the values, but you cannot assume equal steps between them or calculate a meaningful average.

Example. Finishing position in a competition. First place comes before second, and second before third, but the ranks themselves tell you nothing about how far apart the competitors actually were.

An Olympic weightlifting event makes this concrete, because athletes are ranked by a weight, so we can see both scales at once. The weights in Figure 3 are invented.

On the podium, second place looks exactly as far from first as it does from third. On the scale below, second place is 2 kg behind first and 29 kg ahead of third. Ranking keeps the order and throws away the distances.

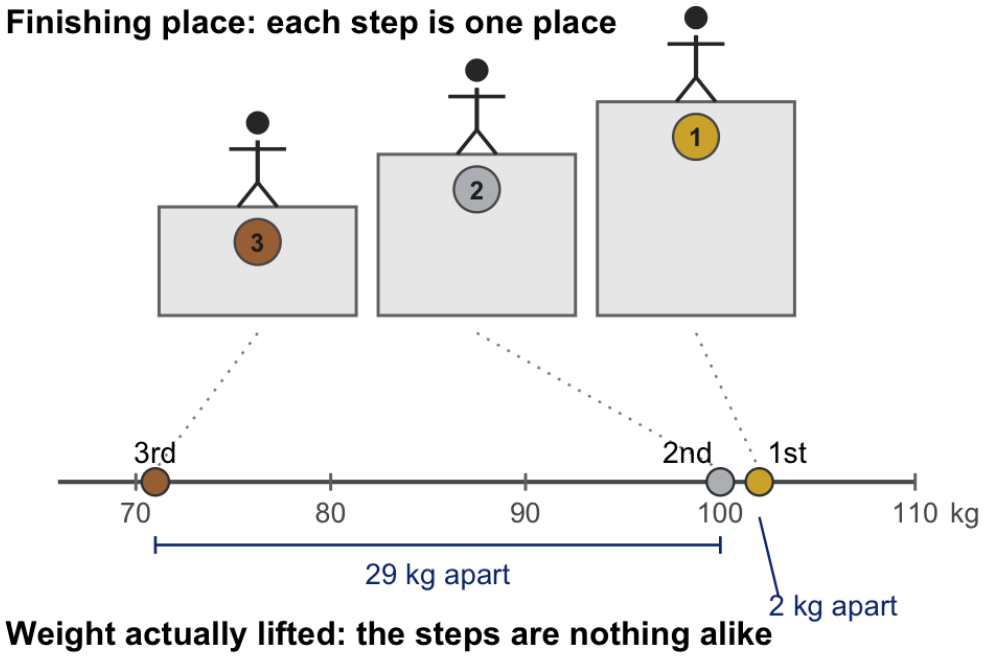


Figure 3: The same three athletes shown two ways. On the podium, each step is one place above the last. On the scale below, the weights they lifted are 29 kg and 2 kg apart.

A single event also gives us both kinds of variable. The finishing place is ordinal, the weight lifted is on a ratio scale (defined below), and only one of them can be averaged.

Interval scale

An **interval** variable has equal spacing between values, so differences are meaningful. Its zero does not mean none of the quantity, so ratios between values are not meaningful. For example, 80°F is not twice as hot as 40°F.

Example. Temperature in degrees Celsius.

A difference of 3° means the same regardless of whether it is 7 → 10 or 15 → 18. But 0°C does not mean "no temperature." That zero is defined by the freezing point of water, which is a useful convention rather than a fact about heat. So you can say today is 3° warmer than yesterday, but you cannot say 20°C is "twice as hot" as 10°C.

Another example. Calendar years. The ten years from 1990 to 2000 are the same length as the ten years from 2010 to 2020, so differences work fine. But the starting point of the count is a convention we agreed on, and other calendars start their count somewhere else and are equally correct. Because the zero is a convention, the year 2000 is not "twice as late" as the year 1000.

Ratio scale

A **ratio** variable has equal spacing between values and a true zero, where zero means none of the quantity. The true zero makes ratios between values meaningful. A 40-year-old tree is twice as old as a 20-year-old one, and 10 kg is twice 5 kg.

Example. Age in years. Zero means no age at all, and someone who is 20 really is twice as old as someone who is 10. Differences (20 - 10 = 10 years) and ratios (20 ÷ 10 = 2) are both meaningful.

Temperature makes an unusually clean comparison here, because the same physical quantity can be recorded on either kind of scale depending on where you put the zero. Figure 4 shows both.

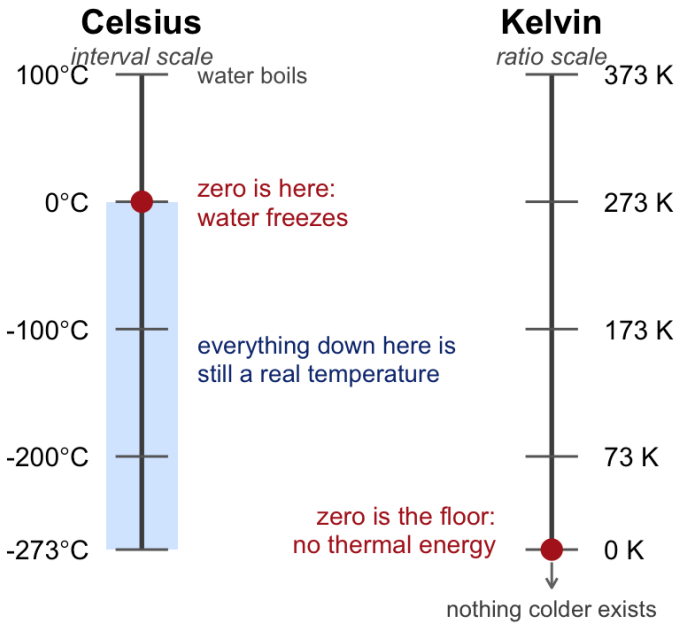


Figure 4: The same physical range of temperatures on two scales. The scales differ only in where they put zero. Celsius anchors its zero to the freezing point of water, and Kelvin anchors it to the absence of thermal energy.

Going from 10°C to 20°C doubles the Celsius number, so it is tempting to call it twice as hot. On the Kelvin scale those same two temperatures are 283 K and 293 K. The increase is 10 K, and 10 / 283 is about 0.035, a physical increase of about 3.5%. The doubling was an artifact of where Celsius happens to put its zero.

Kelvin has a true zero, so its ratios hold. A gas molecule at 200 K has, on average, twice the kinetic energy of one at 100 K.

The table below summarizes which operations each scale supports.

	can rank	can subtract/add	can form ratios	example
nominal				species
ordinal	x			finishing place
interval	x	x		temperature (°C)
ratio	x	x	x	age

Continuous versus discrete variables

Another useful distinction is whether a variable can take on values in between others.

- A **continuous variable** can, in principle, take on any value within a range. For example, consider height. If you are 72 inches tall and your friend Cameron is 71 inches tall, Alan could be 71.4 inches and David 71.49 inches. Because we can always imagine a new value in between two others, height is continuous.
- A **discrete variable** has separate, distinct values with nothing in between. Nominal variables are always discrete, because no species falls "between" a green frog and a brook trout the way 71.4 falls between 71 and 72. Ordinal variables are also discrete. Second place falls between first and third, but nothing can fall between first and second.

Interval and ratio variables can be either. Among ratio variables, height is continuous, but the number of people living in a household is discrete, since you cannot have 4.2 people. Among interval variables, temperature in degrees Celsius is continuous, but the year you started college is discrete, since there is no year between 2022 and 2023.

The table below shows how the scales of measurement relate to this distinction. Cells with an "x" mark what is possible.

	continuous	discrete
nominal		x
ordinal		x
interval	x	x
ratio	x	x

Treating ordinal scales as interval

These categories are guides, not hard rules. For example, survey responses on a 1–5 "strongly disagree" to "strongly agree" scale are technically ordinal, but researchers often treat them as "quasi-interval" because they assume the steps between responses are roughly equal. Treating the responses as quasi-interval lets researchers average them, which the ordinal scale alone does not allow.

Independent and dependent variables

In most studies we have many variables, but when we analyze them, we usually split them into two roles, the thing we're trying to explain and the thing doing the explaining.

The **independent variable (IV)**, X, is the variable used to explain another, and the **dependent variable (DV)**, Y, is the variable being explained. When several variables do the explaining, we number them X_1, X_2, and so on. The logic is that if there's a relationship, Y depends on X. These terms can be confusing for two reasons. IVs are rarely "independent of everything else." Fertilizer amount, for example, may track how much a farmer spends on irrigation. And if there's no relationship, the DV doesn't "depend" on the IV at all.

Two alternative vocabularies are often clearer. In experiments, IVs are **manipulations** and DVs are **measured outcomes**. When we are using X to predict Y rather than manipulating anything, the pair is **predictor** and **outcome**.

role of the variable	classical name	in experiments	in prediction
"to be explained" (Y)	dependent variable (DV)	measured outcome	outcome
"to do the explaining" (X)	independent variable (IV)	manipulation	predictor

> **Note: Which terms this course uses**
>
> You should recognize all three pairs, since all three appear in the literature. This book mostly uses IV and DV. In regression and ANCOVA (Chapters 9 and 11), it also calls any variable used to predict the DV a predictor, whether measured or categorical. Both names refer to the same role.

Experimental and non-experimental research

Research studies are either experimental or non-experimental, and the difference is how much control the researcher has.

In **experimental research**, the researcher deliberately manipulates something (the manipulation, or IV) and measures its effect (the measurement, or DV). An experiment needs three things: the manipulation, a control or comparison group that does not get it, and random assignment of units to the groups. The goal is to isolate causal effects.

The risk is that something other than the manipulation also differs between the groups and affects the outcome. To keep other factors from explaining the result, researchers try to hold them constant. In practice, it's almost impossible to identify *everything* that might matter, much less keep it constant. The standard solution is **randomization**, assigning each unit (each plot, animal, or person in the study) to a condition by chance. Randomization does not remove those other factors. Because chance decides which units go to which condition, any other factor is as likely to end up in one group as in the other, so the groups are less likely to differ systematically.

Some experiments cannot ethically be run. Suppose we wanted to know if smoking causes lung cancer. Observing smokers and non-smokers can only get us so far, because those groups differ in many ways besides smoking, like occupation, income, and diet. A true experiment would require randomly assigning people to smoke or not, and that would be unethical. The same problem limits medical research, where we know surprisingly little about how certain drugs or exposures affect pregnant people, precisely because we cannot ethically assign them to risky conditions.

In environmental science the limitation is usually not ethics but feasibility. There is no control planet to set aside as a baseline, and nobody can change precipitation or temperature across a whole region. Because of limits like these, much of the field relies on **non-experimental research**, research in which the researcher observes conditions instead of assigning them. This chapter covers three common forms.

Quasi-experiments compare conditions the researcher did not create. You cannot control what a factory discharges into a river, but you can compare reaches above and below the outfall and use statistical tools to account for the other factors you could not hold constant.

Time series follow one system over a long stretch, which matters because many environmental processes unfold across years or decades. Tracking global temperature or sea level takes years of consistent measurement, and statistics separates the trend from the natural year-to-year variation.

Case studies go deep on a single event or place: the aftermath of a wildfire, the ecology of a threatened habitat, the consequences of a policy. One case is tied to its context, but a well-designed one can still expose a mechanism that will generalize elsewhere. A drought in one forest can show how water potential and transpiration respond to stress, which helps anticipate what other forests will do.

Table 1 sets the four designs side by side. Only the experiment assigns conditions at random, which is why it supports the strongest claims about cause.

Table 1: Four kinds of study compared on who sets the conditions, whether they are randomized, and what each can say about cause.

Design	Who sets the conditions?	Ran-dom-ized?	Environmental example	What it can say about cause
Experiment	The researcher	Yes	Fertilizer plots	Strongest
Quasi-experiment	Nature or someone else	No	Reaches above vs. below an outfall	With care, after accounting for other factors
Time series	Nobody (one system over time)	No	Global temperature record	Trend, not cause
Case study	Nobody (one event)	No	One forest's drought response	A mechanism to test elsewhere

From measurement to description

The first half of this chapter was about what data *are*: how a broad concept gets operationalized into a variable, what scale that variable is on, and what kind of study produced it. The rest of the chapter is about what to do with a dataset once you have one.

> *Far better an approximate answer to the right question, which is often vague, than an exact answer to the wrong question, which can always be made precise.*
>
> John W. Tukey

Statistics often begins with too much data. Nobody can learn much by reading rainfall records from hundreds of stations or millions of satellite pixels one number at a time. We need tools to summarize and visualize. **Descriptive statistics** are the numbers and plots that summarize a dataset. They describe the data you have. Later chapters turn to inferential statistics, which use a sample to draw conclusions about a larger population.

There is more than one useful way to describe a dataset, and which one you use depends on the pattern you want to show. In the rest of this chapter, a plot of every daily temperature shows the seasonal pattern, a histogram shows which temperatures were common, and single numbers like the mean and the range summarize the center and the spread.

Looking at the data

From here on we work with a real dataset.

R ships with `airquality`, a record of daily air quality measurements taken in New York from May through September of 1973. Each row is one day, with ozone concentration (parts per billion), solar radiation, wind speed, and maximum temperature. Table 2 shows the first six days.

Table 2: The first six days of the airquality dataset.

Ozone	Solar.R	Wind	Temp	Month	Day
41	190	7.4	67	5	1
36	118	8.0	72	5	2
12	149	12.6	74	5	3
18	313	11.5	62	5	4

Table 2: The first six days of the airquality dataset.

Ozone	Solar.R	Wind	Temp	Month	Day
NA	NA	14.3	56	5	5
28	NA	14.9	66	5	6

Look at the `NA` entries. `NA` is R's code for a missing value, and here it means ozone and solar radiation were not recorded every day. Ozone is missing on 37 of the 153 days, and real environmental datasets almost always have gaps like this. In R, `airquality$Ozone` picks out the Ozone column of the `airquality` data, and `mean()` averages it:

```
mean(airquality$Ozone)
#> [1] NA
```

R returns `NA` rather than a number, because it has no way of knowing what the missing days would have been. You have to tell it explicitly to set those days aside:

```
mean(airquality$Ozone, na.rm = TRUE)
#> [1] 42.12931
```

`na.rm = TRUE` tells R to remove the `NA` values before computing, and you will use it throughout the course.

Scatter of raw values

Plot every value. Before summarizing anything, look at the data.

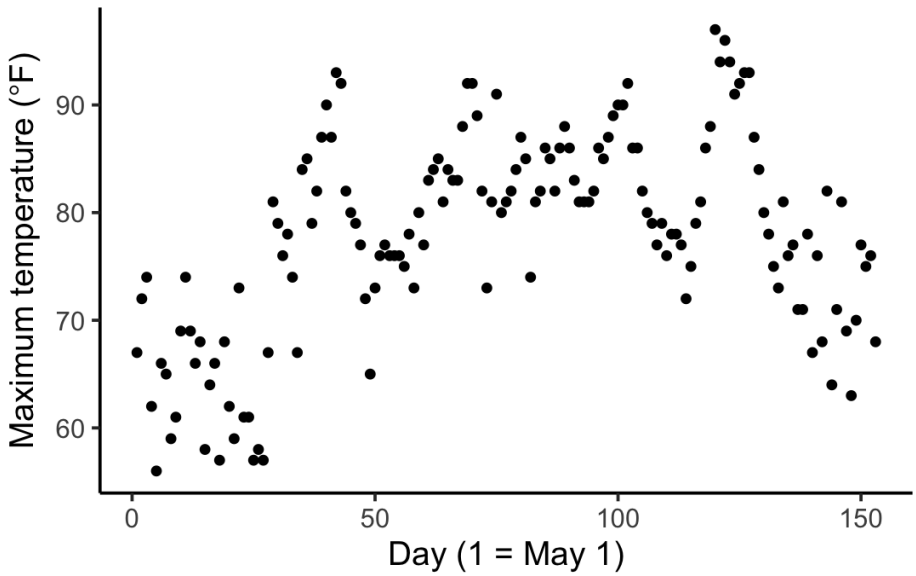


Figure 5: Daily maximum temperature at a New York station, May through September 1973.

Figure 5 shows all 153 daily temperatures at once, with day number across the bottom and that day's maximum temperature up the side. This already tells us more than a table of 153 numbers would. There

is a clear seasonal arc, with daily highs mostly in the 60s in May and in the 80s through July and August, then cooling again by September.

One question this plot answers poorly is whether most days were warm or cool. To answer it here, you end up squinting at a cloud of dots, because the seasonal ordering gets in the way. For that we need a different view, a histogram.

Histograms

A **histogram** summarizes data by grouping values into bins rather than showing each point separately. A bin is a range of values along the horizontal axis, and each bar's height is the number of values that fall in that bin. It throws away the day-to-day ordering on purpose, and in exchange it shows you the shape of the data. Figure 6 shows the same daily temperatures as a histogram.

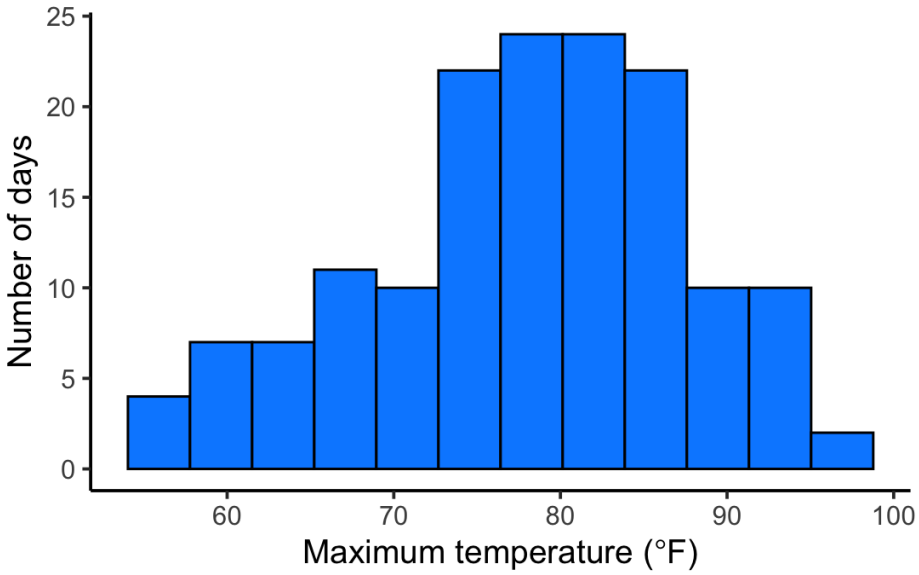


Figure 6: A histogram of the same daily temperatures.

Each bar counts the days falling in a temperature range. A histogram looks like a **bar chart**, a graphical display that summarizes the frequency of categorical data, but the two are not the same. In a bar chart each bar is a separate category, so the bars could go in any order. In a histogram the bars are bins on a numeric scale, so they sit side by side in order. Most days fall in the 70s and 80s, with a tail of cooler days down to 56°F and only 14 days above 90°F.

Bin width is a choice you make. Wide bins look smooth but hide detail. Narrow bins show more variation, but with only a few days in each bin the bar heights jump around at random. The bins divide the same temperature range, so choosing the number of bins sets their width, and fewer bins means wider bins. Figure 7 compares the same temperatures at 5, 10, 20, and 50 bins.

All four show the same general shape, but the 5-bin and 50-bin versions are hard to read. Five bins is so coarse that the peak becomes two wide blocks. Fifty spreads 153 days so thinly that the bars jump around at random. The 10-bin and 20-bin versions are the ones you would use.

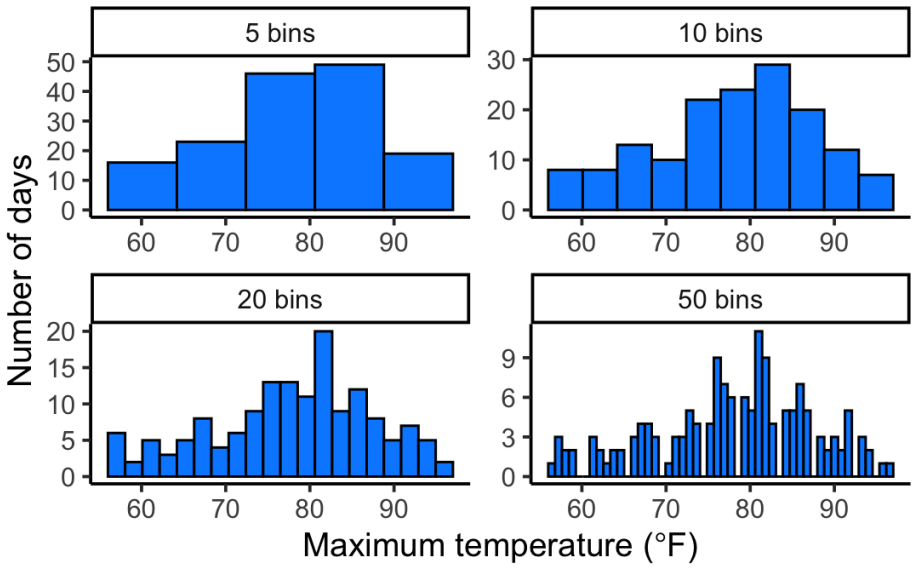


Figure 7: Four histograms of the same temperature data using different numbers of bins.

Distribution, central tendency, and variation

Three terms will keep coming up: distribution, central tendency, and variation. Their everyday meanings aren't far from the statistical ones.

- A **distribution** is the pattern of which values occur in a dataset and how often each occurs. A histogram is one way of showing a distribution. The shape of the distribution tells us where values are concentrated, where they're rare, and how far the extremes reach. Later we'll also use distributions to generate theoretical data, which is the basis for sampling distributions (Chapter 3) and tools like *t*-tests (Chapter 6).
- **Central tendency** is a single representative value near the center of a distribution, summarizing what most of the data are like. It is about *sameness*, where values cluster. In the temperature histogram, the tallest bars are in the high 70s and low 80s, so we say the data have a central tendency around 80°F. Central tendency doesn't mean "everything is the same", just that many values gather in a common area. Some datasets even have more than one cluster, and therefore more than one central tendency.
- **Variation** is how spread out the values in a dataset are. It describes how the values differ, their *differentness*. If every day had been 78°F, there would be no variation at all. Instead the days run from 56°F to 97°F, so there is variation to describe. Variation is measured by a family of statistics. This chapter introduces the first of them, the range, and the next chapter builds the variance and the standard deviation.

Variation is one feature of a distribution. The distribution is the whole pattern of values, which includes where they cluster, how far they spread, and the shape they make.

Measures of central tendency (sameness)

A measure of central tendency summarizes a dataset with a single number for where its values cluster, its "center." The plot of every value and the histogram show where values fall, but they do not give that number.

The temperature histogram shows a peak somewhere around 80°F, but not one value you could report or compare with another dataset.

Mode

The **mode** is the most frequently occurring value: count how often each value appears, and whichever wins is the mode. In `1 1 1 2 3 4 5 6` the mode is 1, since it appears three times.

Ties are allowed. In `1 1 1 2 2 2 3 4 5 6` both 1 and 2 appear three times, so the data have two modes.

The mode can be far from the center. In `1 1 2 3 4 5 6 7 8 9` the mode is 1, but eight of the ten values are larger.

The mode is most useful when values in a dataset repeat, as they do in counts or in categories like species.

Median

The **median** is the exact middle of the data once they are ordered from smallest to largest. For example:

1 5 4 3 6 7 9

Ordered from smallest to largest, they are:

1 3 4 5 6 7 9

The median is 5, with three numbers on each side.

With an even count there is no single middle value, so the median is the midpoint of the two central numbers. In `1 2 3 4 5 6` that is halfway between 3 and 4, or 3.5.

An extreme value moves the median far less than the mean. The mean, defined in the next section, is the sum of the values divided by how many there are. Consider `1 2 3 4 4 4 5 6 6 6 7 7 1000`. Most values are modest and one is wildly out of line. The median is 5, in the middle of the bulk of the data. The 13 values sum to 1055, so the mean is 1055 / 13 = 81.15. If that last value were 8 instead of 1000, the sum would be 63 and the mean would fall to 63 / 13 = 4.85. The median would still be 5, because the seventh of the 13 ordered values is still 5.

That value of 1000 is an outlier. An **outlier** is a value much farther from the bulk of the data than the rest of the values. Outliers can change the mean a great deal while changing the median very little, and how to handle them is a question we return to later in the course. Table 3, in the section comparing the three measures, sets these numbers beside the same comparison for the ozone data.

Mean

The **mean**, also called the average, is the sum of a set of numbers divided by how many numbers there are.

$$\text{Mean} = \bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

The $\sum$ symbol is a capital Greek **sigma**, and it means "sum these up." (The lowercase σ, which you will meet later as a standard deviation, means something different.) The x_i are the individual numbers, i runs from the first to the last, n is how many there are, and $\bar{x}$ is the mean. In plainer terms, it is the sum of your numbers divided by the count of your numbers.

For the five numbers 3, 7, 9, 2, and 6, that gives

$$\bar{x} = \frac{3+7+9+2+6}{5} = \frac{27}{5} = 5.4$$

Like the mode, the mean is a good summary for some data and a poor one for others. Later in this chapter, a few very high ozone days raise the mean about 10 ppb above the median.

What does the mean mean?

The mean formula divides a sum by a count. To see what that represents, start with plain division. When we compute $\frac{12}{3} = 4$, we are splitting 12 into three equal parts of 4 each. Division splits the numerator into equal pieces.

The same logic applies to the mean. Suppose you add up every day's rainfall across the summer into one total. If that total were spread equally across all the days, each day would get the same amount. That amount is the mean. If five days had 3, 7, 9, 2, and 6 mm of rain, the total of 27 mm spread over five days gives 5.4 mm each. Individual days were wetter or drier than that.

The mean is the only single value that can replace every observation and still leave the total unchanged.

Figure 8 shows this for the five numbers from earlier. The parts of the bars above the mean of 5.4 add up to 5.8 (1.6 + 3.6 + 0.6 for the values 7, 9, and 6), and the gaps below it also add up to 5.8 (2.4 + 3.4 for the values 3 and 2). Moving the excess into the gaps levels every bar at the mean.

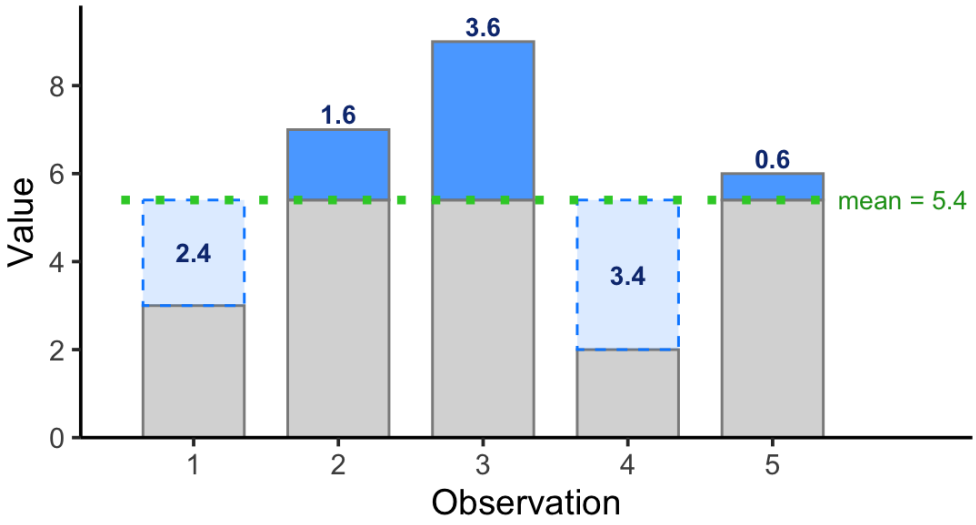


Figure 8: The five values 3, 7, 9, 2, and 6 with their mean of 5.4 as a dotted line. The shaded parts above the line exactly fill the outlined gaps below it.

The weighted mean

When you average across groups that aren't the same size, a plain average of the group means does not give the mean of all the observations. A variant of the mean, the weighted mean, does give the mean of all the observations. We introduce it here because it returns later in the course.

Suppose you're monitoring vegetation recovery at three restored wetland sites. Each site was surveyed with a different number of plots:

Site	Mean vegetation cover (%)	Plots (n)	Plots × mean
A	46	12	552
B	58	14	812
C	73	30	2190
Total		56	3554

We want the average cover across all 56 plots. The tempting answer is to average the three site means:

$$\frac{46 + 58 + 73}{3} = \frac{177}{3} = 59$$

That answer treats a 12-plot site and a 30-plot site as equally informative, when Site C has more than twice as many observations as Site A. A **weighted mean** multiplies each value by a weight w_i, which sets how much that value counts, adds the products, and divides by the sum of the weights. It fixes this problem:

$$\bar{x}_w = \frac{\sum_i w_i x_i}{\sum_i w_i} = \frac{\text{weighted sum}}{\text{total weight}}$$

Here the values being averaged are the site means, so we index sites with j, and the natural weight for a group mean is that group's sample size, $w_j = n_j$:

$$\bar{x}_w = \frac{\sum_j n_j \bar{x}_j}{\sum_j n_j} = \frac{(46)(12) + (58)(14) + (73)(30)}{12 + 14 + 30} = \frac{3554}{56} = 63.5$$

The weighted answer, 63.5, is 4.5 points above the unweighted 59, because Site C, with the highest mean of 73, has 30 of the 56 plots.

You have already met this idea outside of statistics. A course grade works the same way, since an exam worth 60% counts three times as much as a quiz worth 20%. The formula is the same, with percentages as the weights instead of sample sizes.

Whenever you see an average of averages, check whether the groups behind it were the same size.

Comparing the mean, median, and mode

So far we have three summaries of the "middle" of a dataset. When they disagree, the disagreement tells you about the shape of the distribution.

A distribution is **symmetric** when its left and right halves are mirror images around the center. It is **right-skewed** when a long tail of high values stretches out to the right, and **left-skewed** when the long tail is on the left.

Temperature's mean (77.9°F) and median (79°F) are only about 1°F apart, which is what you expect when a distribution is roughly symmetric. Ozone's mean (42.1 ppb) and median (31.5 ppb) are about 10 ppb apart.

For a right-skewed distribution you will usually find mode < median < mean, which is exactly what Figure 9 shows: the mode at 23 ppb, the median at 31.5 ppb, and the mean at 42.1 ppb.

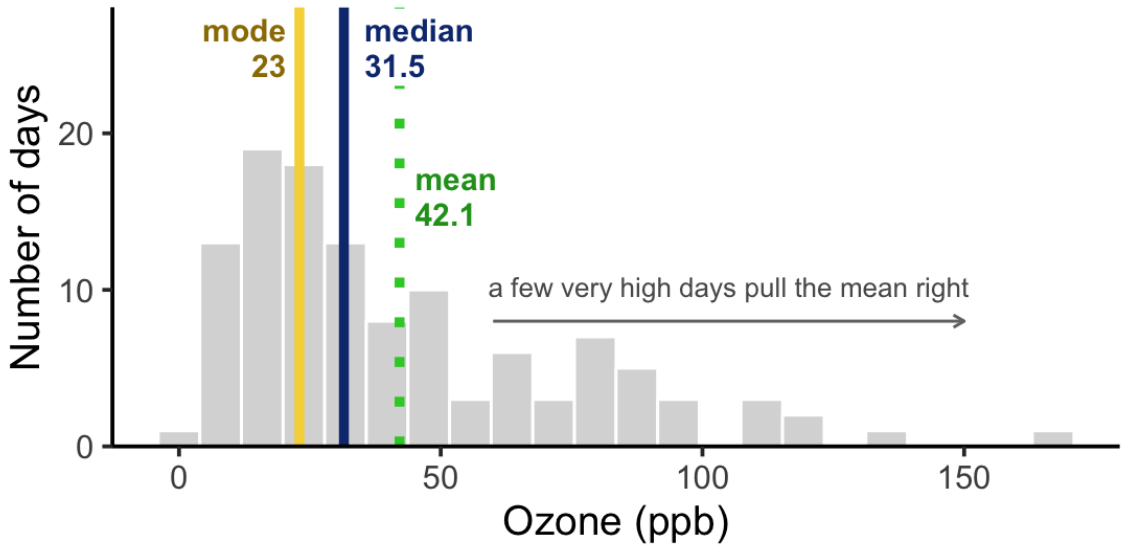


Figure 9: Daily ozone concentration with all three measures of central tendency marked. They fall in the order mode, median, mean, which is the usual order for a right-skewed distribution.

The mode line is not at the tallest bar, because the mode counts exact repeats while a bar counts every day within its range. 23 ppb occurred 6 times, and no other single value occurred more than 4.

The three measures differ because each one is sensitive to something different.

- The mean uses every value, so the seven bad-air days above 100 ppb raise it. Add one extreme day and the mean moves.
- The median depends only on which value is in the middle. Adding one value at the high end moves the middle position up by half a place, so the median moves at most to a neighboring value, however extreme the new day is.
- The mode ignores position entirely and reports whichever value occurred most often, so it tells you what happened most.

In a right-skewed distribution, the long tail of high values pulls the mean up the most and the median up a little, while the mode stays at the most common value near the peak. That gives the order mode $<$ median $<$ mean.

Table 3 puts numbers on the first two points, using the small dataset from the median section and the ozone days. In the small dataset the median stays at 5 while the mean falls from 81.15 to 4.85 when 1000 is replaced by 8. Dropping the 7 high ozone days moves the median from 31.5 to 29 ppb, and the mean from 42.1 to 36.8 ppb.

Table 3: How much one extreme value, or a few extreme days, moves the median and the mean.

Data	Median	Mean
1 2 3 4 4 4 5 6 6 6 7 7 8	5	4.85
1 2 3 4 4 4 5 6 6 6 7 7 1000	5	81.15
Ozone, all 116 days	31.5	42.1
Ozone, without the 7 days above 100 ppb	29	36.8

The mean and the median both describe the ozone data correctly, and they answer different questions. If

you want to know what a typical day looked like, the median is the better guide. If you care about the total ozone burden across the summer, the mean is that total divided by the number of days.

Table 4 lists which scales of measurement each measure of center works with, and Table 5 matches each distribution shape to the measure that best describes a typical value.

Table 4: Which scales each measure of center works with.

Measure	Works with	Affected by extreme values?
Mean	Interval, ratio	Yes
Median	Ordinal, interval, ratio	Very little
Mode	Nominal, ordinal, interval, ratio	No

Table 5: Which measure of center best describes a typical value for each distribution shape.

Shape	What you will see	Best measure of a typical value	Example
Roughly symmetric	mean ≈ median ≈ mode	Mean	Daily maximum temperature (Figure 6)
Right-skewed	mode < median < mean	Median	Daily ozone (Figure 9)
Left-skewed	mean < median < mode	Median	Exam scores on an easy test
Bimodal	Two peaks, with the mean and median between them where few values fall	The two modes, reported separately	Two plant varieties measured together

Warning: When the mode gets fragile

Ozone is a continuous measurement, so exact repeats are largely an accident of rounding. The value 23 ppb occurs 6 times out of 116 days, barely more often than several neighbors. Round differently and the mode moves. It is most useful for categorical variables, where "most common species" is exactly the question, and for values that repeat exactly.

Measures of variation (differentness)

Central tendency tells us what values have in common. Measures of variation tell us how they differ. Any dataset whose values are not all identical shows some variation, and summarizing that spread matters as much as finding the center, because two datasets with the same center can differ widely in spread. Daily highs that all fall between 76°F and 80°F and daily highs that run from 56°F to 97°F could share a mean near 78°F.

The range

Consider these 10 ordered numbers:

1 3 4 5 5 6 7 8 9 24

The **range** is the largest value minus the smallest value. Here the smallest is 1 and the largest is 24, so the range is 24 - 1 = 23. The range quickly shows the boundaries of the data and can flag possible outliers. For instance, if you expected values between 1 and 7 but found one at 340,500, you'd know something unusual happened and might investigate.

The range uses only two numbers and ignores everything in between, so a single unusual value moves it a long way. Drop the single value of 24 and the largest value becomes 9, so the range falls from 23 to 9 - 1 = 8. We want a measure built from *every* observation, one that says how far a typical value is from the center.

That measure is the variance, and its square root the standard deviation. We build both in the next chapter, because the same building block, how far each value is from the mean, describes how one variable varies and also how two variables vary *together*, which is what correlation measures.

Remember to look at your data

Descriptive statistics reduce a dataset to a few numbers, so they always lose detail. That's fine if the summary captures the important features, but sometimes it doesn't.

> Tip: Always pair a summary with a plot
>
> Compute the descriptive statistics, then plot the data, and check that the two tell the same story.

We just paired a summary with a plot for ozone. The 10 ppb gap between the mean and the median suggested a right skew, and the histogram confirmed it.

When the numbers and the graph disagree, believe the graph. A summary can only report what it was designed to measure, while a plot also shows features you did not think to measure. The next chapter makes this concrete with a famous set of four datasets that share nearly every summary statistic and look nothing alike.

Chapter Summary

Why it matters. Every analysis later in this course assumes you already have numbers that mean something and a sense of what they look like. This chapter covers where data come from and how to describe them once you have them.

Core ideas

- **Why statistics matters.** It provides a disciplined way to separate a real effect from ordinary variation and from our own expectations.
- **Operationalization and measurement.** Turning a broad concept like *soil health* into something recordable is a decision you have to make and justify, not something the data hand you.
- **Scales of measurement.** Nominal, ordinal, interval, and ratio, plus the continuous-versus-discrete distinction. These determine which statistical tools are available to you. Chapter 7 uses the ordinal scale to choose a rank-based test over a t-test.
- **Independent and dependent variables.** The roles variables play in an analysis, and the several vocabularies used to describe them.
- **Experimental and non-experimental research.** Environmental science rarely permits full experimental control, so the field relies on quasi-experiments, long-term time series, and case studies.
- **Plot first.** A plot of every value shows patterns such as a seasonal arc, a histogram shows the shape of the distribution, and bin width is a choice that changes what you see.
- **Central tendency.** The mode, median, and mean each summarize the "middle" differently, and they diverge when data are skewed.

- **The weighted mean.** When you average group means from groups of unequal size, weight each group mean by its group's n. A plain average of group means treats a small group as being as informative as a large one.
- **Variation.** The range gives a first, blunt sense of spread. The measures built from every observation, the variance and the standard deviation, come in the next chapter alongside correlation, since all three are built from deviations from the mean.
- **Always look at the graph.** A summary reports only what it was built to measure, so pair every one with a plot.

Stevens, S. S. 1946. "On the Theory of Scales of Measurement." *Science (New York, N.Y.)* 103: 677–80. https://doi.org/10.1126/science.103.2684.677.